Physical Succession and Eulerian Electromagnetic Identity

Positive transport, exact orbit reduction and response through moving distributions

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Abstract

We construct physical distribution progression through positive source histories, reciprocal collision receiving, exact conditional orbit bundles, moving-background response and an Eulerian reference chart. These operations retain the current, charge, work and torque generated by the same progressing participants. Exact half-turn symmetry halves the independently advanced histories on its admitted carrier. The reference construction expresses distribution transport while material labels circulate. Together they provide executable tools for source-renewed plasma research and a precise source-to-exit equation for the next resident-population construction.

1. Response through a moving physical organization

For the nonlinear field-material-alpha-support law $\dot{z} = \mathcal{F}(z)$, the finite-time response follows

$$\dot{V}(t) = D\mathcal{F}(z_0(t))V(t), \qquad V(0) = I. \tag{D2}$$

The background is the actual nonlinear progression. Field, density, flow, stress and projected current provide physical Eulerian readouts. Individual alpha coordinates are not required to return when they circulate through a stationary distribution.

Atlas XI advances paired nonlinear perturbations and checks the differentiated composition at rebased states. At 32 ns, halving the response step from 250 ps to 125 ps changes the magnetic directional response by 0.0659026% [C24]. Independent one-nanosecond chain-rule tests at several progressing backgrounds agree in field and alpha-source readouts. This is an executable finite-time moving response. The admitted background is not periodic, so its result is not a Floquet certificate.

Where a relative-periodic state is later constructed, the appropriate return derivative is

$$\mathcal{M}_T = D(\mathcal{R}_{-\Omega T} \circ \Phi_T)(z_*). \tag{D3}$$

Eliminating explicit orbit coordinates can produce a causal memory term and an initial-condition contribution. Those terms cannot be replaced by an instantaneous fitted gain without changing the operation. The retained moving-response code provides the law against which such a future reduction can be tested.

2. Exact half-turn orbit bundles

On the admitted even-harmonic field, with $Q = \mathbf{diag}(-1, -1, 1)$, equivariance gives

$$\mathcal{P}_t(Qx, Qv) = Q\mathcal{P}_t(x, v). \tag{D4}$$

The same symmetry must hold for sources, collision law and boundary response. Under those conditions 1,096 independently advanced representatives reconstruct 2,192 physical histories. Weights remain positive and endpoints remain the actual propagated states. Work and torque are accumulated with the corresponding partner. If an operative odd harmonic breaks the symmetry, the quotient must be released. This is a constructive cost reduction with a precise admission condition.

For finite orbit averages,

$$\bar{\mu}_N = \frac{1}{N}\sum_{k=0}^{N-1} \mathcal{P}_T^k \mu_0, \qquad \mathcal{P}_T\bar{\mu}_N - \bar{\mu}_N = \frac{\mathcal{P}_T^N\mu_0 - \mu_0}{N}. \tag{D5}$$

The identity is telescopic. Its usefulness depends on the topology and source/work readouts in which the endpoint term is controlled. It does not automatically turn finite orbital averaging into a full collisional stationary population.

3. Positive source-age progression

Let S be the actual birth measure and $\mathcal{P}_a$ the positive characteristic/collision/exit propagator. The young occupation is

$$\mu_T = \int_0^T \mathcal{P}_a S\, da. \tag{D6}$$

The continuing resident population satisfies

$$f_* = \mu_T + \mathcal{P}_T f_*. \tag{D7}$$

For a supplied, time-homogeneous sub-Markov operator with sufficient eventual removal and finite occupation, the positive Neumann sum $\sum_{k\geq 0} \mathcal{P}_T^k \mu_T$ is the corresponding stationary occupation. Finite residence and the required field/work moments must actually be established. A matrix inverse alone is not a physical source-to-exit proof.

The program constructs six-dimensional source histories, physical age prefixes, independent one-period characteristics and one-period nuclear/collision renewal actions. The collision receivers carry species momentum and energy. The alpha current and charge are deposited from the same measure. Continuing old-energy cohorts are retained; they are not converted into fictitious ash when a sparse representation cannot hold them.

For a rigidly rotating supplied field, the Jacobi relation is $K = H - \Omega P_\phi$. Collisionless preservation of K permits H and P_ϕ to change together. Uniform redistribution on a Jacobi leaf can therefore introduce energy and torque that no physical characteristic supplied. The earlier implied mixing obligation is retained as a rejected representation, not a source of useful wave power.

4. Eulerian distribution transport through a reference chart

Let Φ_t^0 be an invertible reference flow and suppose its continuous reference measure is invariant:

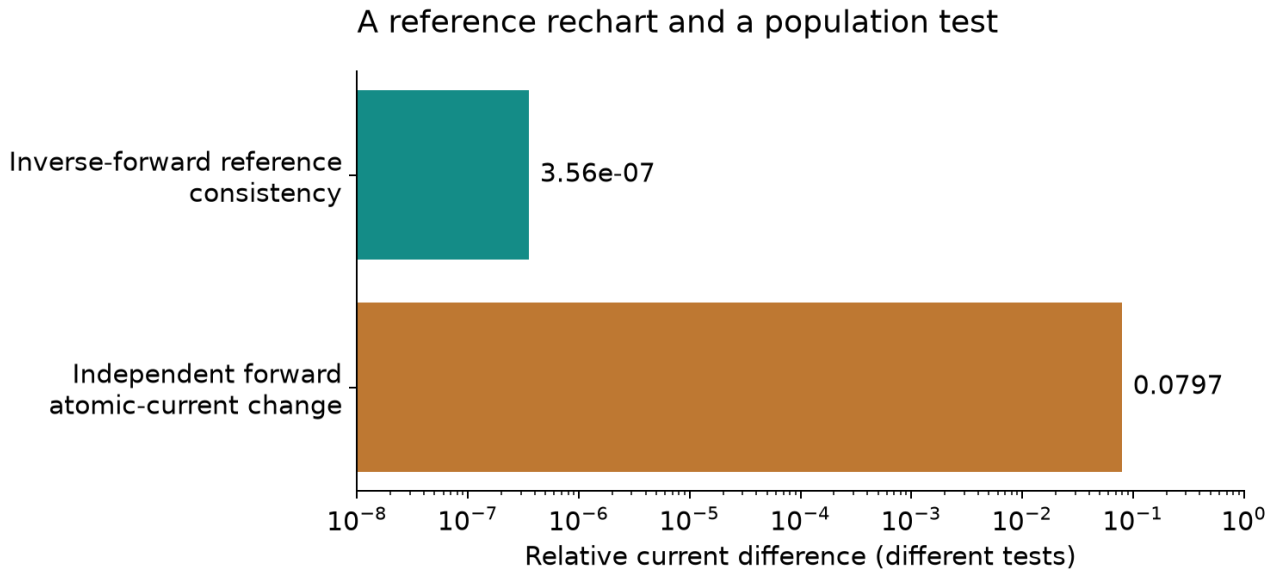
$$\Phi_{t\#}^0 \mu_0 = \mu_0. \tag{D8}$$

Then for an engaged physical flow Φ_t,

$$\Phi_{t\#}\mu_0 = (\Phi_t \circ \Phi_{-t}^0)_{\#}\mu_0. \tag{D9}$$

The proof uses pushforward composition and D8. This is a coordinate representation of the same transported measure. The inverse-reference path is bookkeeping, not a physical energy supplier. If the reference is sourced, absorbing or collisional, invertibility and invariance require separate treatment; D9 cannot silently supply them.

The numerical inverse-forward check reaches current consistency 3.56285×10^{-7} and charge consistency approximately 2.31×10^{-8} on its refined one-period characteristic operation [C21]. The independent forward test of the inherited finite atomic reference has a 7.9735% current change [C22]. This measures a different relation. It neither invalidates D9 under D8 nor proves that an arbitrary smooth invariant distribution must change. It shows why a cancellation identity cannot certify the admitted source population.



Two distinct reference tests. A small inverse-forward defect does not replace an independent population-invariance test.

5. Physical histories and positive collision transport

Sparse six-dimensional endpoint snapping produced large artificial work and torque. A strict repair retained unresolved physical endpoints; its substantial unresolved throughput prevented a false stationary claim. Whole-path and moment compression likewise have to be tested against their next field-source response. These failures identify where a representation lacks an actual continuation map.

The positive alternative is to preserve physical histories, use exact admitted symmetries, and construct a reduction only where the next transport and readouts factor through it. Positive collision cubature is an example: its children are actual velocity states with weights, drift and diffusion moments and complementary material receiving. Repeated branching, however, is not yet a validated long-lifetime compression.

6. One-period nuclear and collision receiving

The one-period renewal action includes actual D/T depletion, alpha and neutron products, species collision receiving and electromagnetic work. Its positive deterministic diffusion cubature retains all generated children. Energy, count and canonical accounts are checked with compensated summation where large populations make naive subtraction misleading.

These calculations evaluate the physical renewal residual on a supplied field. They do not solve the coupled nonlinear root with a stationary full-age source. Nor do the between-event nonlinear moving runs silently contain continuous births or collisions. The archived README identifies exactly which source operations each experiment admits.

The interior ellipsoidal event level 0.99 is a numerical admission margin, not the material surface at level one. Its crossings must not be counted as physical losses in D7. The near-sheet continuation comparison is a separate field-map diagnostic. Preserving that distinction keeps both the useful transport operation and its boundary account meaningful.

7. The compositional condition supporting these operations

Operational freedom. Researchers can advance physical source measures, calculate their electromagnetic response, exploit an actual orbit symmetry, and change distribution coordinates through a derived reference map. The constructed object is the progressing field-coupled population. Its numerical labels are a means of carrying that operation.

From Closure to Composition §3.1 states an exact condition. If equal signatures preserve admitted engagements, required outputs and renewed signatures, an update can act through that signature:

$$r \circ U_\eta = \bar{U}_\eta \circ r. \tag{D1}$$

For transport this means more than reproducing density or current at one instant. The representation must retain what is needed for the next trajectory, deposited source, work, torque and exit operation. The D-Series distinction between common provenance and actual succession supplies the dependency ground. The identity is the compositional condition supporting the transport operations constructed here.

8. The available distribution-level operations

What this opens. The source measure can be advanced through actual birth, characteristic and receiving operations; exact symmetry can lower orbit cost; a conditional reference chart can express the same transported measure; and the field response can follow a moving background. These tools make distribution-level control and resident renewal concrete research problems with executable progression maps.

These progressing maps supply a concrete research instrument: nonlinear differentiation, causal moving response, positive histories, exact orbit bundles, reciprocal receiving and a conditional Eulerian rechart. Use demonstration sets `moving`, `orbit`, `renewal` and `reference` to execute their distinct operations. Their next integrated composition is the resident fixed point solved with its field, material and source state.

Source navigation: Bibliography · demonstration recipes.